NAME

saw − function

LIBRARY

ciaaw library (*libciaaw*, *−lciaaw*)

SYNOPSIS

Fortran **function saw(***s***,***ab***,***u***)result(res)**

character(len=*), intent(in) :: *s*

logical, intent(in), optional :: *ab*

logical, intent(in), optional :: *u*

real(dp) :: *res*

C **double ciaaw_saw(char ***s*,**int** *n*,**bool** *ab*,**bool** *u***);**

Python **saw(***s:***str,***abridged:***bool=True,***uncertainty:***bool=False)->float:**

DESCRIPTION

Get the standard atomic weight for the element *s*. The default behavior is to provide the abridged value,*ab is TRUE*, recommended by the CIAAW. The uncertainty can be retrieved by setting *u*=TRUE.

Arguments:

s	Element symbol.
ab	Set to False if the abridged value is not desired. Default to TRUE.
u	Set to True if the uncertainty is desired. Default to FALSE.

RETURN VALUE

NaN	If the provided element is incorrect.
-1	If the element does not have a SAW.

EXAMPLE

Fortran:

```
use ciaaw
saw("H", ab=.false., u=.true.)
```

C:

```
include <stdio.h>
include "ciaaw.h"
ciaaw_saw("H", 1, false, true)
```

Python:

```
import pyciaaw
pyciaaw.saw("H", ab=False, u=True)
```

SEE ALSO

ciaaw(3), **ciaaw_cli**(1), **ciaaw_version**(3), **ciaaw_saw**(3), **ciaaw_ice**(3), **ciaaw_nice**(3), **ciaaw_naw**(3), **ciaaw_nnaw**(3)